1.0 COURSEWORK TITLE

APU Automotive Service Centre (APU – ASC)

2.0 THE COURSEWORK OVERVIEW

The assignment is to design and implement a system to support APU Automotive Service Centre (APU – ASC). There are 4 types of end users interacting to the APU – ASC:

1. Managers
2. Counter Staff
3. Technicians
4. Customers

The application should have the following features:

- Login accesses
- User registrations
- Set prices
- Create new appointments
- Assign new appointments
- Check and update assigned appointments
- Collect payments and generate receipts
- Access service and payment histories
- Provide feedbacks and comments
- Analysed reports

In addition, a supporting document is needed to reflect the design of the implementation codes and the implementation details that utilize the Object-oriented programming concepts.

3.0 OBJECTIVES OF THIS COURSEWORK

Develop the practical ability to describe, justify, and implement an Object-oriented system.

4.0 TYPE

Group Assignment (minimum 3 and maximum 4 students)

5.0 COURSEWORK DESCRIPTION

As an object-oriented programming student, you are required to identify the relationship among the entities and develop a user-friendly application for different types of users by fulfilling the basic features that listed in section 2.0.

The application should be designed with four types of access rights such as managers, counter staff, technicians, and customers. Table 1.0 shows the functionalities that can be accessed by each user upon login.

Roles	Functionalities
Managers	❖ Create/Read/Update/Delete managers, counter staff, and technicians. ❖ Set prices of normal service and major service. ❖ View all feedbacks and comments. ❖ Analyzed reports.
Counter Staff	❖ Edit personal / individual profile. ❖ Create/Read/Update/Delete customers. ❖ Create and assign new appointments (normal service 1 hour and major service 3 hours) to available technicians. ❖ Collect payment and generate receipts.
Technicians	❖ Edit personal / individual profile. ❖ Check comments and details of individual assigned appointments. ❖ Update individual assigned appointments to "Completed". ❖ Provide feedbacks for individual assigned appointments.
Customers	❖ Edit personal / individual profile. ❖ Access individual service and payment histories. ❖ Access feedbacks of individual appointments. ❖ Provide comments for counter staff and technicians involved in individual appointments.

Table 1.0

6.0 General Requirements

- The system submitted should compile and be executed without errors.
- User input validation should be done to avoid logical error.
- The system must run continuously.
- The implementation code must highlight the use of Object-oriented programming concepts.
- Students should use text files for storing and retrieving data required for the system.
- Database tools like access, oracle, sql servers, etc. are not allowed.

- Graphical user interface must be included by using libraries in java.awt and java.swing packages.

7.0 Deliverables

- Report and source code
- Submission deadline: 29th May 2026, 11:59pm

7.1 Report:

The report needs to be submitted in softcopy to Moodle with:

A. Cover Page
   - Module
   - Coursework Title
   - Intake code
   - Student name and id
   - Date Assigned (The date the report was handed out)
   - Date Completed (The date the report is due to be handed in)

B. Table of Contents

C. Contents
   - Design solution (Use Case Diagram and Class Diagram)
   - Screenshots of output of the program with appropriate explanations
   - Description and justification of Object-oriented concepts incorporated into the solution
   - Additional feature

D. Limitation and Conclusion

E. References

F. Appendix
   - Screenshots of all text files that are used to store the data

7.2 Source Code:

Application source code need to be compiled into zip folder and submitted to Moodle.

8.0 Assignment Assessment Criteria

The assignment assessment consists of two main components: System Implementation (40%) and System Documentation (15%).

Assessment	Description	Weightage (%)
System Implementation	Performance result of all operations and design per requirements	16
	Appropriate design and the implementation of codes illustrating the object-oriented programming concept incorporated	16
	Presentation	8
System Documentation	Description and justification of the object-oriented concepts incorporate	6
	UML Diagrams	4
	Program output screenshots	3
	Report format and references	2

9.0 Development Tools

The program must be written in Java language, and you can use any Java development IDE as a tool, but back-end data must be stored in **.txt** files.

10.0 Academic Integrity

You are expected to maintain the utmost level of academic integrity during the duration of the course. Plagiarism is a serious offence and will be dealt with according to regulations of Asia Pacific University on plagiarism.